

SIMATSENGINEERING(SAVEETHASCHOOLOFENGINEERING)
DepartmentofComputerScienceandEngineering

ASSESSMENTTOOL2-ConceptMapping

CourseCode:	CSA0305	CourseTitle:	DataStructures
COAssessed:	CO1-Imitation(BL3,BL4)	Assessment:	AssessmentTool2-Concept Mapping
Weightage:	35%ofCIA	TotalMarks:	100
CO1Statement:	ApplysuitabledatastructuresandADTsforefficientproblemsolvingbyanalyzingtimeandspace complexity.		
Student Name:	M.VAISHNAV	Reg. No.:	192524251
Section/Batch:		Date:	10/07/2026

CodeofConduct

I M.VAISHNAV certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: M.VAISHNAV (192524251)

Instructions

- Thistestcontains5Conceptmappingquestions.Total:100Marks.
- Whenastudentanswerusingaconceptmap,inevaluationtheanswersshouldbewithfollowingfour requirements: Understanding of concepts, Connections between ideas, Logical structure, Clarity of representation.
- Nonenegativemarking.Unansweredquestionscarryzeromarks.

Section/Topic	Focus	QNos	Marks
Linkeddata structure	Basicdata structure	1	20
PrimitiveandNon-PrimitiveDataStructure-Linear and Non-Linear Data Structure	ClassificationsofLineardatastructure	2	20
SearchData Structure	LinearSearchandBinary Search	3	20
Recursion,Performance analysis	Recursion	4	20
TreeData Structures	ClassificationsofTreedatastructure	5	20
GRANDTOTAL:		100Marks	

Annexure A – Concept Mapping

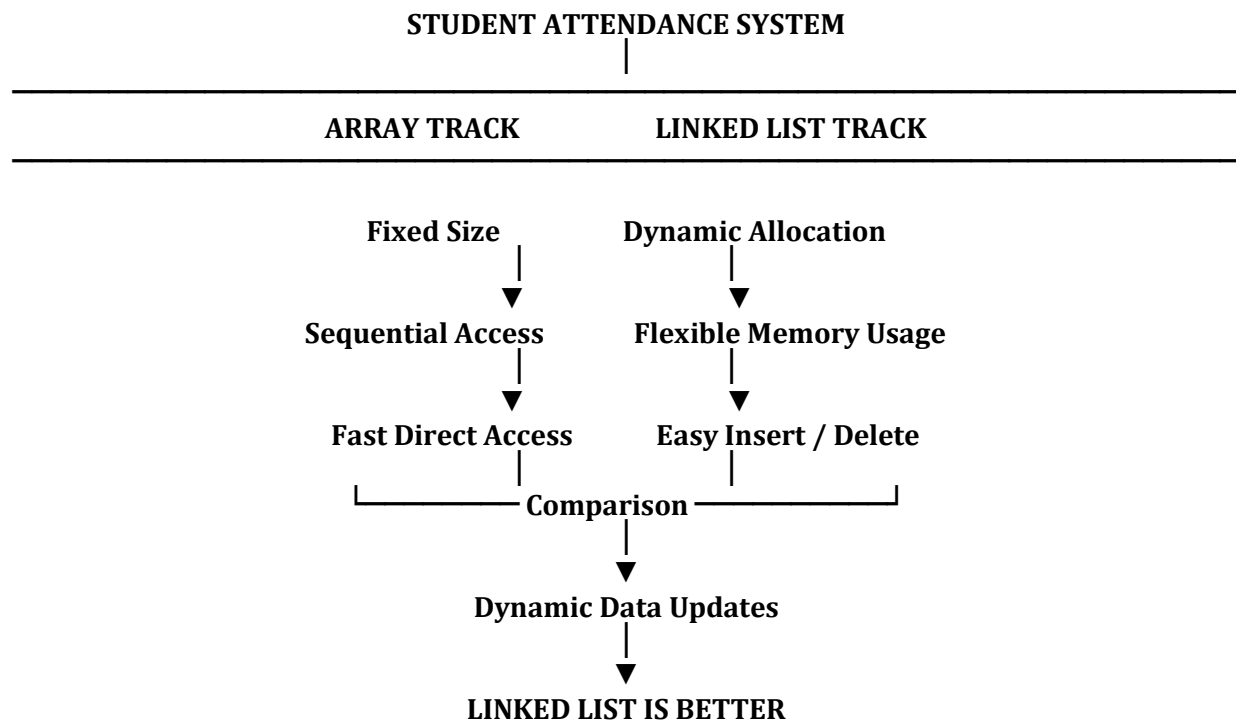
1. A student attendance system records daily presence data. Construct a concept map linking Array → Fixed Size → Sequential Access → Linked List → Dynamic Allocation → Flexibility. Explain how each structure works and justify which is more suitable for dynamic data updates. (20 Marks)
2. A weather monitoring system collects temperature, humidity, and wind data continuously. Construct a concept map linking Primitive Data Types → Non-Primitive Structures → Linear Structures → Non-Linear Structures → Applications. Include examples and explain how each structure supports continuous data processing. (20 Marks)
3. A digital archive stores millions of documents. Construct a concept map linking Searching → Linear Search → Binary Search → Sorted Data → Time Complexity → Performance. Analyze the impact of data organization on search efficiency and justify the best technique. (20 Marks)
4. A program processes nested expressions using recursion and may encounter deep nesting. Construct a concept map linking Recursion → Call Stack → Stack Overflow → Error Handling → Efficiency. Analyze potential issues and justify strategies to handle them. (20 Marks)
5. A company maintains hierarchical employee records with multiple levels of management. Construct a concept map connecting Tree → Nodes → Levels → Traversal (DFS/BFS) → Searching → Efficiency. Analyze how traversal techniques affect performance and justify the most suitable approach for retrieving employee data. (20 Marks)

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding; some interpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

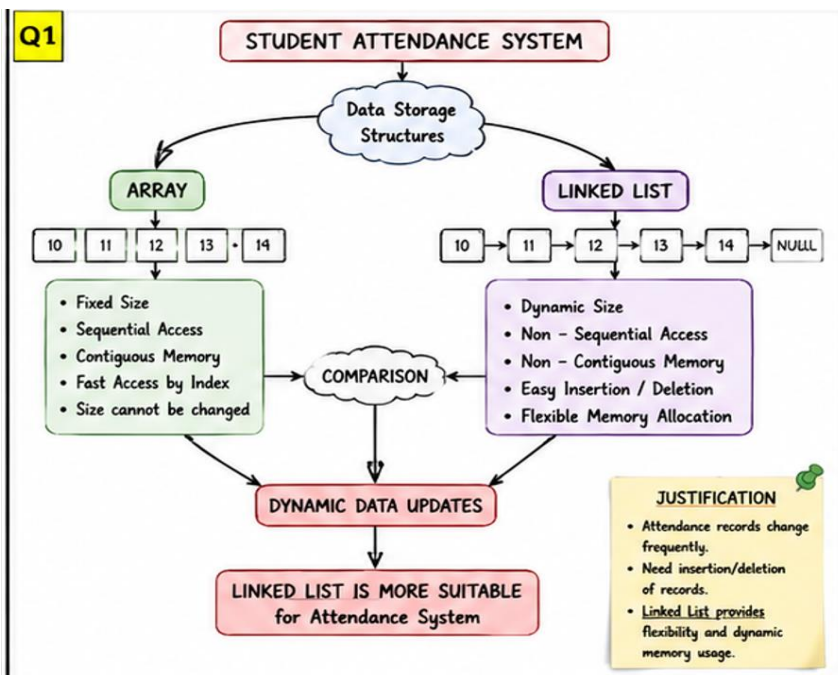
Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Q1. STUDENT ATTENDANCE SYSTEM – RAILWAY TRACK CONCEPT MAP



Explanation:

- Array stores data in contiguous memory locations.
- Array size cannot be changed after creation.
- Linked List stores data dynamically using nodes.
- Linked List allows insertion and deletion easily.
- Attendance records change frequently.
- Therefore, Linked List is more suitable for dynamic updates.



Linked Data Structure – Student Attendance System

Understanding of Concepts

- Arrays store elements in contiguous memory locations.
- Linked Lists store data using nodes connected through links.
- Both are linear data structures used for data storage.

Connections Between Ideas

- Attendance records need frequent insertion and deletion.
- Arrays provide fast access but fixed size.
- Linked Lists provide dynamic memory allocation and flexibility.

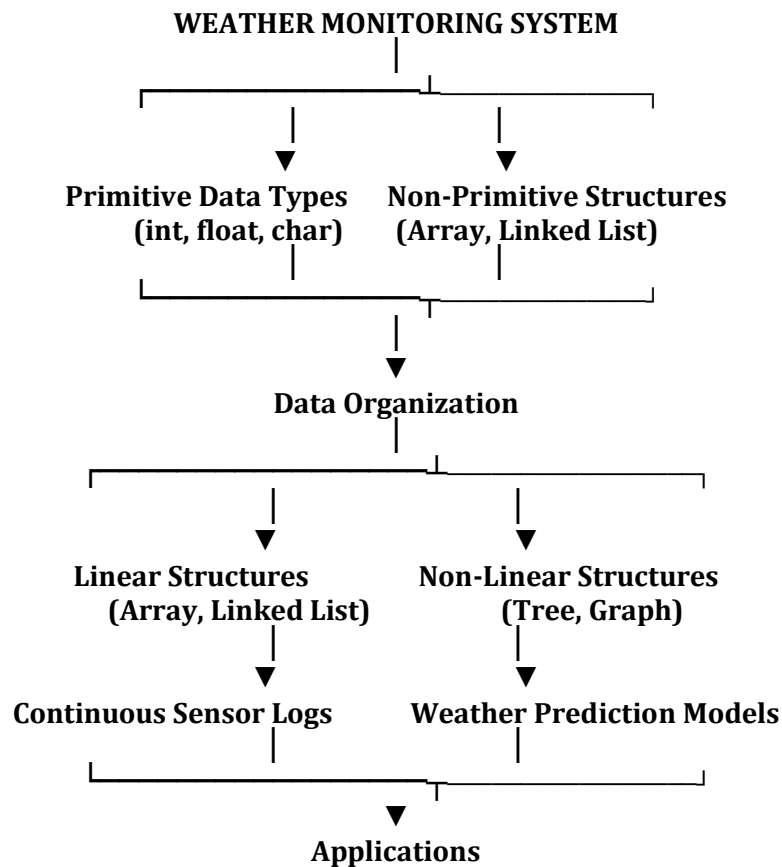
Logical Structure & Justification

- Compare Array and Linked List features.
- Analyze their advantages and limitations.
- Conclude that Linked List is more suitable for attendance management because records change frequently.

Clarity of Representation

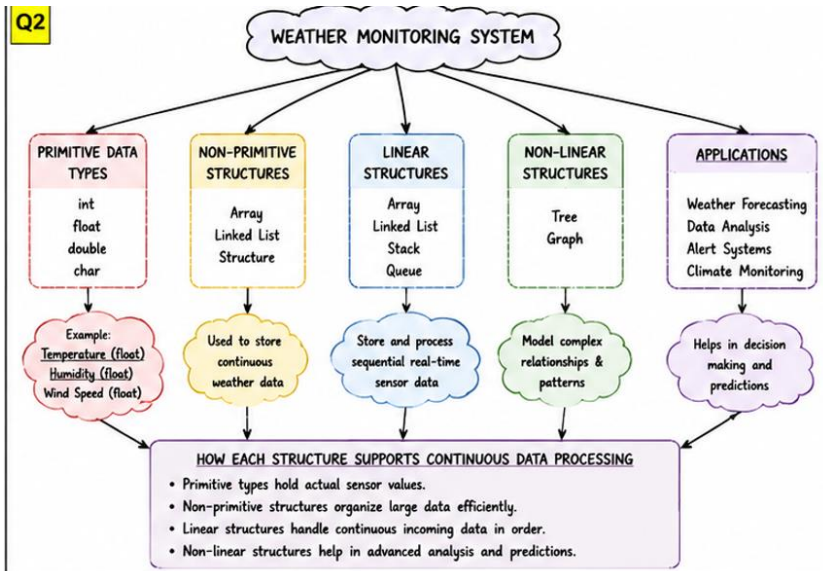
- Railway-track concept map clearly separates both structures.
- Arrows show comparison and decision-making process.
- Key features are easy to understand visually.

Q2. WEATHER MONITORING SYSTEM – TREE CONCEPT MAP



Explanation:

- Primitive types store temperature, humidity, and wind speed.
- Arrays and Linked Lists store continuous sensor readings.
- Trees and Graphs help in complex weather analysis.
- Linear structures process sequential data efficiently.
- Non-linear structures support advanced forecasting systems.



Primitive and Non-Primitive Data Structure – Weather Monitoring System

Understanding of Concepts

- Primitive data types include int, float, char, and double.
- Non-primitive structures include arrays, linked lists, trees, and graphs.
- Data structures help organize weather information efficiently.

Connections Between Ideas

- Primitive types store sensor values such as temperature and humidity.
- Linear structures manage sequential weather data.
- Non-linear structures support prediction and analysis.

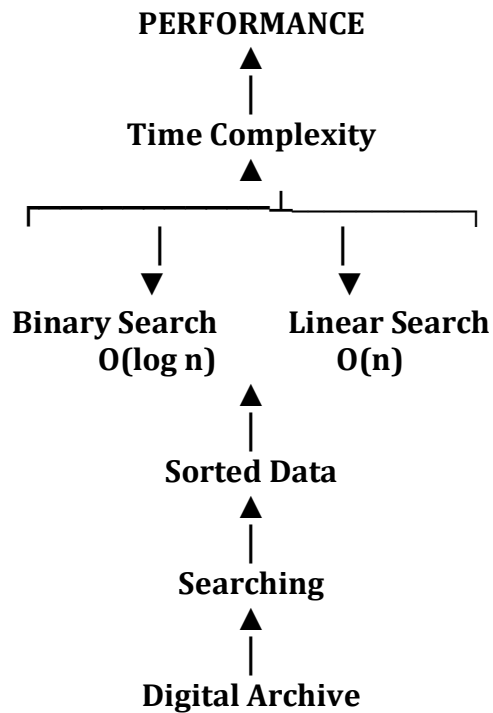
Logical Structure & Justification

- Start with data classification.
- Connect primitive and non-primitive structures.
- Show how each structure contributes to weather monitoring and forecasting.

Clarity of Representation

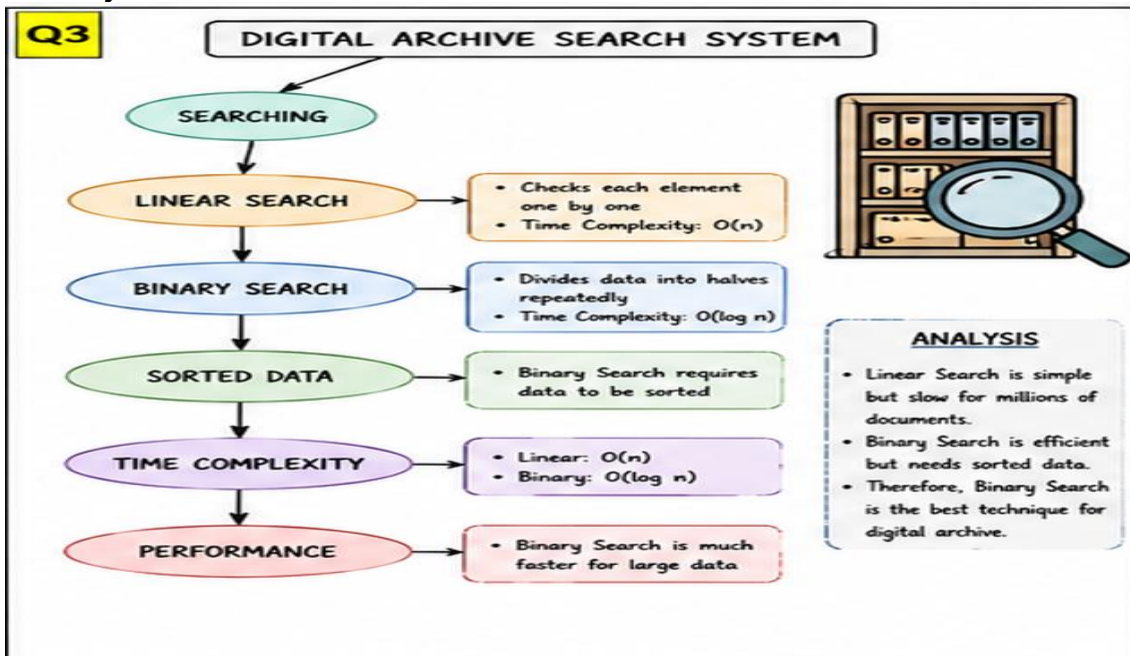
- Tree-style concept map clearly displays hierarchy.
- Branches indicate relationships among structures.
- Easy visualization of data classification.

Q3. DIGITAL ARCHIVE – PERFORMANCE PYRAMID CONCEPT MAP



Explanation:

- Linear Search checks every document one by one.
- Binary Search divides data into halves repeatedly.
- Binary Search requires sorted data.
- Time complexity of Linear Search is $O(n)$.
- Time complexity of Binary Search is $O(\log n)$.
- Binary Search is the best choice for millions of documents.



Search Data Structure – Digital Archive Search

Understanding of Concepts

- Searching helps locate records efficiently.
- Linear Search examines elements one by one.
- Binary Search repeatedly divides sorted data.

Connections Between Ideas

- Binary Search requires sorted data.
- Time complexity influences performance.
- Faster search methods improve archive retrieval efficiency.

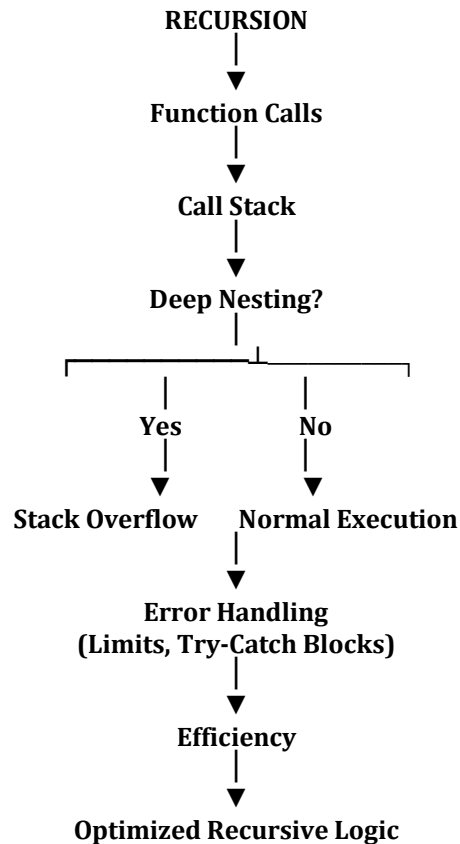
Logical Structure & Justification

- Compare Linear Search and Binary Search.
- Analyze time complexity $O(n)$ and $O(\log n)$.
- Justify Binary Search as the best option for large digital archives.

Clarity of Representation

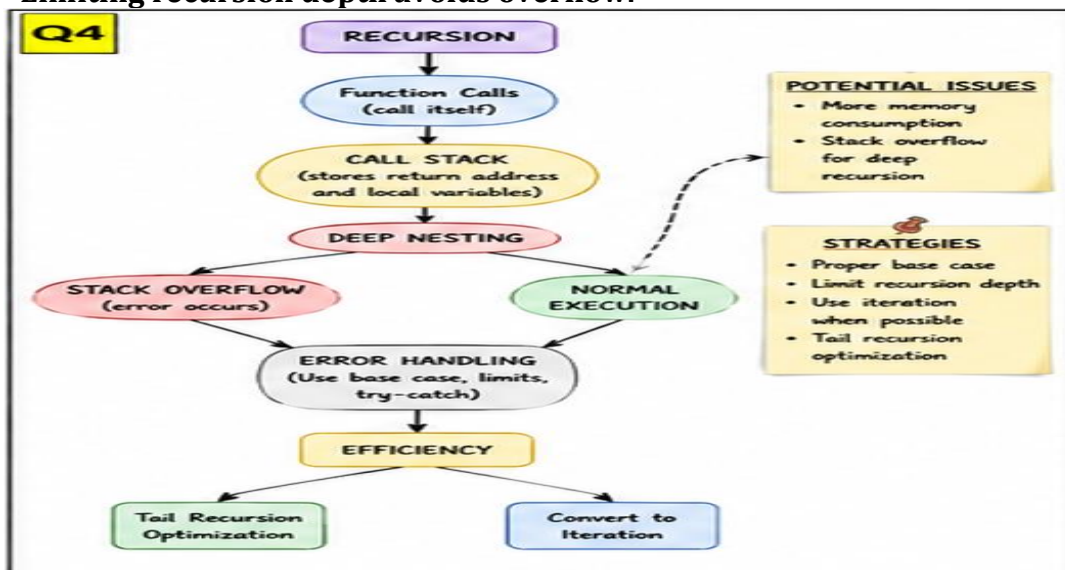
- Pyramid structure emphasizes performance comparison.
- Information flows from searching to efficiency.
- Visual hierarchy makes concepts easy to follow.

Q4. RECURSION - WATERFALL CONCEPT MAP



Explanation:

- Recursion solves problems by calling itself.
- Every recursive call occupies stack memory.
- Excessive recursion can cause stack overflow.
- Error handling prevents program crashes.
- Tail recursion and iteration improve efficiency.
- Limiting recursion depth avoids overflow.



Recursion and Performance Analysis

Understanding of Concepts

- Recursion is a technique where a function calls itself.
- Recursive calls use stack memory.
- Excessive recursion may cause stack overflow.

Connections Between Ideas

- Function calls create a call stack.
- Deep recursion increases memory usage.
- Error handling and optimization improve performance.

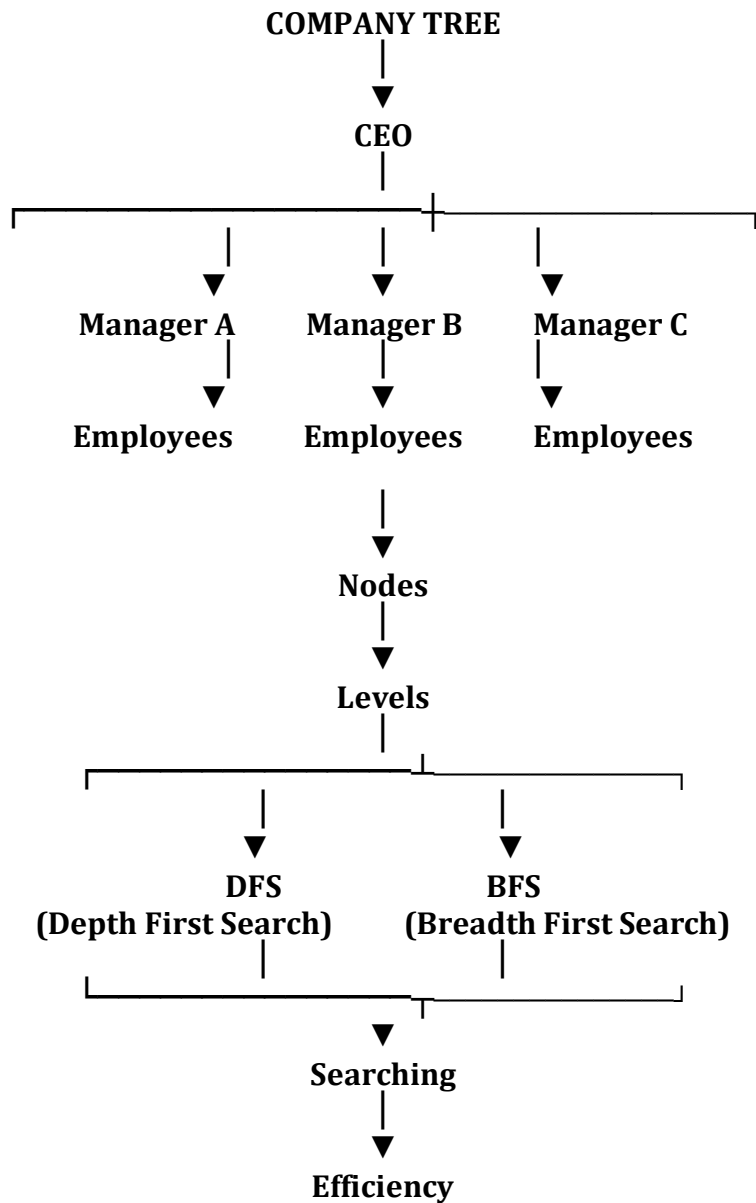
Logical Structure & Justification

- Begin with recursion process.
- Show problems such as stack overflow.
- Present solutions like tail recursion and iteration.

Clarity of Representation

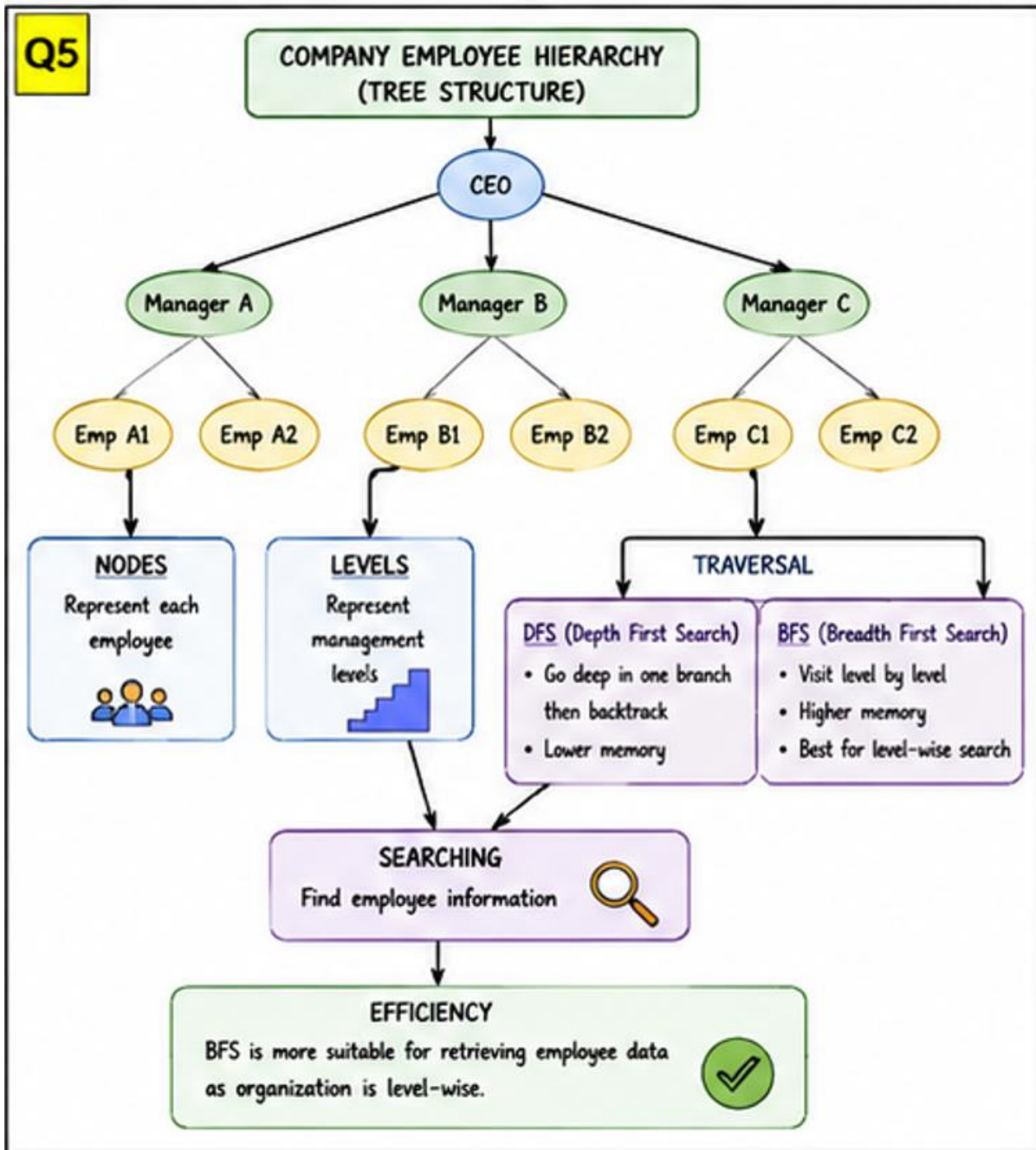
- Waterfall concept map shows step-by-step execution.
- Cause-and-effect relationships are clearly visible.
- Easy to understand recursion flow.

Q5. EMPLOYEE MANAGEMENT SYSTEM – ORGANIZATION CHART CONCEPT MAP



Explanation:

- Employee hierarchy is represented using a Tree.
- Each employee is a node.
- Management layers form levels.
- DFS explores one branch completely before moving.
- BFS explores level by level.
- BFS is best for retrieving employee information level-wise.

Q5

Tree Data Structure – Employee Hierarchy

Understanding of Concepts

- Trees are non-linear hierarchical data structures.
- Nodes represent employees.
- Levels represent management hierarchy.

Connections Between Ideas

- CEO acts as the root node.
- Managers and employees form branches.
- DFS and BFS are traversal methods used for searching.

Logical Structure & Justification

- Represent organization as a tree.
- Compare DFS and BFS traversal techniques.
- Conclude BFS is more suitable for level-wise employee retrieval.

Clarity of Representation

- Organization-chart layout resembles real company hierarchy.
- Nodes and levels are clearly identified.
- Traversal methods are easy to compare visually.